Sample Program Guide

Revision. 2.6

November 14, 2022

Introduction

This guide explains how to use this sample program.

Usage

Supported camera

D5500, D5600

Environment

The sample program is command prompt, runs on these environments.

For the operating environment and development environment of the sample program, refer to the following chapter of ReadMe_Eng.txt.

- Environment of operation
- Contents - [Windows] - Sample Program
- Contents - [Macintosh] - Sample Program

Files

The sample program needs following files. Copy these files on the appropriate folder and use it.

Windows

File Name	Folder
Type0016.md3	Same as this sample program
NkdPTP.dll	Same as Type0016.md3
dnssd.dll	Same as this sample program
NkRoyalmile.dll	Same as this sample program

- If you want to change the folder of Type0016.md3, you should modify Seach_Module() function of Function.cpp.

Macintosh

File Name	Folder
Type0016 Module.bundle	Same as this sample program
libNkPTPDriver2.dylib	SYSTEM/Library/Application Support/Nikon/Camera Control Modules/
Royalmile.framework	Same as libNkPTPDriver2.dylib

- If you want to change the folder of Type0016 Module.bundle, you should modify Search_Module() function of Function.cpp.
- The meaning of SYSTEM is the directory where OS is installed.

Main Menu

When you start the sample program, it loads module and opens module object. Then main menu is shown.

1) **Select Device**

If some devices are connected to PC, you can choose which device will be used. Then sub menu 1 is shown.

2) **AsyncRate**

Show a recommended interval of Command_Async.
(refer to 9.1 kNkMAIDCapability_AsyncRate in MAID3.PDF)

3) **IsAlive**

Show the status of Module object.
(refer to 9.6 kNkMAIDCapability_IsAlive in MAID3.PDF)

4) **Name**

Show the name of Module object.
(refer to 9.9 kNkMAIDCapability_Name in MAID3.PDF)

5) **ModuleType**

Show the type of Module object.
(refer to 9.53 kNkMAIDCapability_ModuleType in MAID3.PDF)

6) **Version**

Show version of MAID supported by the module.
(refer to 9.57 kNkMAIDCapability_Version in MAID3.PDF)

Sub Menu 1

You can read or write the settings of the source object.

1) **Select Item Object**

If some pictures are stored in camera, you can choose which item object will be used. Then sub menu 2 is shown. (When this menu returns the message “There is no item” though Capture command is succeeded, please retry this menu until new item object is detected.)

2) **Camera settings(1)**

You can change some camera settings. Sub menu 3 will be shown.

3) **Camera settings(2)**

You can change other camera settings. Sub menu 4 will be shown.

4) **Shooting Menu**

You can change photography settings. Sub menu 5 will be shown.

5) **Live View**

You can change Live View settings. Sub menu 6 will be shown.

6) **Custom Menu**

You can change custom settings. Sub menu 7 will be shown.

7) **Async**

Send kNkMAIDCommand_Async to detect in camera condition.
(refer to 9.1 kNkMAIDCommand_Async in MAID3.PDF)

8) **CameraType**

Show the type of camera.
(refer to 3.107 kNkMAIDCapability_CameraType in MAID3Type0016.PDF)

9) **Autofocus**

Perform auto focus.
(refer to 4.27 kNkMAIDCapability_AutoFocus in MAID3Type0016.PDF)

10) **Capture**

Take a picture. If you try to transfer images to PC, you must change 21)SaveMedia of Sub Menu 3 to SDRAM. When you set it to Card, the module does not notify of the addition of the item object.
(refer to 4.17 kNkMAIDCapability_Capture in MAID3Type0016.PDF)

11) **TerminateCapture**

Terminate bulb exposure shooting and record to SDRAM the image until stopped bulb.
(refer to 3.118 kNkMAIDCapability_TerminateCapture in MAID3Type0016.PDF)

12) **PreCapture**

Take a picture for presetting white balance.
(refer to 3.96 kNkMAIDCapability_PreCapture in MAID3Type0016.PDF)

Sub Menu 2

You can read or write the settings of the item object.

1) **Select Data Object**

You can choose data object.
If image object is selected, sub menu 8 will be shown. If movie object is selected, sub menu 9 will be shown. If thumbnail object is selected, sub menu 10 will be shown.

2) **Delete**

Delete the DRAM image of Item object.
(refer to 3.101 kNkMAIDCapability_DeleteDramImage in MAID3Type0016.PDF)

3) **IsAlive**

Show the status of this Item object.

(refer to 9.6 kNkMAIDCapability_IsAlive in MAID3.PDF)

4) Name

Show the name of this Item object.

(refer to 9.9 kNkMAIDCapability_Name in MAID3.PDF)

5) DataTypes

Show the data types supported by this Item object.

(refer to 9.12 kNkMAIDCapability_DataTypes in MAID3.PDF)

6) DateTime

Show the date and time when this Item was created.

(refer to 9.13 kNkMAIDCapability_DateTime in MAID3.PDF)

7) StoredBytes

Show the file size of this Item in bytes.

(refer to 9.14 kNkMAIDCapability_StoredBytes in MAID3.PDF)

Sub Menu 3

You can read and write the detailed settings (1) of the source object.

1) IsAlive

Show the status of this Source object.

(refer to 9.6 kNkMAIDCapability_IsAlive in MAID3.PDF)

2) Name

Show the name of this Source object.

(refer to 9.9 kNkMAIDCapability_Name in MAID3.PDF)

3) Interface

Show the physical interface of this Source object.

(refer to 9.11 kNkMAIDCapability_Interface in MAID3.PDF)

4) DataTypes

Show the data types supported by this Item object.

(refer to 9.12 kNkMAIDCapability_DataTypes in MAID3.PDF)

5) BatteryLevel

Show battery level.

(refer to 4.47 kNkMAIDCapability_BatteryLevel in MAID3Type0016.PDF)

6) FlashMode

Set the flash mode of this source.

(refer to 9.52 kNkMAIDCapability_FlashMode in MAID3.PDF)

7) LockFocus

Show the status of focus lock.

(refer to 3.80 kNkMAIDCapability_LockFocus in MAID3Type0016.PDF)

8) LockExposure

Show the status of exposure lock.

(refer to 3.79 kNkMAIDCapability_LockExposure in MAID3Type0016.PDF)

9) ExposureStatus

Show the exposure value measured by camera.

(refer to 3.81 kNkMAIDCapability_ExposureStatus in MAID3Type0016.PDF)

10) ExposureMode

Set the exposure mode of camera.

(refer to 3.62 kNkMAIDCapability_ExposureMode in MAID3Type0016.PDF)

11) ShutterSpeed

Set the exposure time of camera.

(refer to 3.57 kNkMAIDCapability_ShutterSpeed in MAID3Type0016.PDF)

12) Aperture

Set the aperture of camera.

(refer to 3.60 kNkMAIDCapability_Aperture in MAID3Type0016.PDF)

13) FlexibleProgram

Set the flexible program of camera.

(refer to 3.58 kNkMAIDCapability_FlexibleProgram in MAID3Type0016.PDF)

14) ExposureComp

Set the exposure compensation of camera.

(refer to 3.63 kNkMAIDCapability_ExposureComp in MAID3Type0016.PDF)

15) MeteringMode

Set the metering mode of camera.

(refer to 3.61 kNkMAIDCapability_MeteringMode in MAID3Type0016.PDF)

16) FocusMode

Show focus mode.

(refer to 3.84 kNkMAIDCapability_FocusMode in MAID3Type0016.PDF)

17) FocusAreaMode

Set focus area mode.

(refer to 3.66 kNkMAIDCapability_FocusAreaMode in MAID3Type0016.PDF)

18) FocusPreferredArea

Change focusing point.

(refer to 3.59 kNkMAIDCapability_FocusPreferredArea in MAID3Type0016.PDF)

19) FocalLength

Show focal length(unit: mm).

(refer to 3.83 kNkMAIDCapability_FocalLength in MAID3Type0016.PDF)

20) ClockDateTime

Adjust internal clock.

(refer to 3.56 kNkMAIDCapability_ClockDateTime in MAID3Type0016.PDF)

21) SaveMedia

This will set the recording media.

(refer to 3.114 kNkMAIDCapability_SaveMedia in MAID3Type0016.PDF)

Sub Menu 4

You can read and write the detailed settings (2) of the source object.

1) LockCamera

Keep the camera from direct operation(not through host computer).

(refer to 3.106 kNkMAIDCapability_LockCamera in MAID3Type0016.PDF)

2) LensInfo

Show lens information(focal length and aperture).

(refer to 3.92 kNkMAIDCapability_LensInfo in MAID3Type0016.PDF)

3) UserComment

Set characters which will be written in a new picture file.

(refer to 3.50 kNkMAIDCapability_UserComment in MAID3Type0016.PDF)

4) EnableComment

Enable to add UserComment to an image file.

(refer to 3.51 kNkMAIDCapability_EnableComment in MAID3Type0016.PDF)

5) IsoControl

Set whether auto sensitivity control is used when you take a picture.

(refer to 3.19 kNkMAIDCapability_IsoControl in MAIDType0016.PDF)

6) NoiseReduction

Set whether noise reduction is used when you take a picture.

(refer to 3.20 kNkMAIDCapability_NoiseReduction in MAIDType0016.PDF)

Sub Menu 5

You can read and write settings concerned with photography.

1) CompressionLevel

Set compression level of the image file created by camera.

(refer to 3.2 kNkMAIDCapability_CompressionLevel in MAID3Type0016.PDF)

2) ImageSize

Set the compression level of picture.

(refer to 3.1 kNkMAIDCapability_ImageSize in MAID3Type0016.PDF)

3) WBMode

Set white balance mode(auto/Incandescant/…).

(refer to 3.3 kNkMAIDCapability_WBMode in MAIDType0016.PDF)

4) Sensitivity

Set sensitivity of camera.

(refer to 3.4 kNkMAIDCapability_Sensitivity in MAIDType0016.PDF)

5) WBTuneAuto

Adjust white balance in Auto mode.

(refer to 3.6 kNkMAIDCapability_WBTuneAuto in MAIDType0016.PDF)

6) WBTuneIncandescent

Adjust white balance in Incandescent mode.

(refer to 3.7 kNkMAIDCapability_WBTuneIncandescent in MAIDType0016.PDF)

7) WBFluorescentType

Set the fluorescent type when the WBMode is “Fluorescent”.

(refer to 3.8 kNkMAIDCapability_WBFluorescentType in MAIDType0016.PDF)

8) WBTuneFluorescent

Adjust white balance in Fluorescent mode.

(refer to 3.9 kNkMAIDCapability_WBTuneFluorescent in MAIDType0016.PDF)

9) WBTuneSunny

Adjust white balance in Sunny mode.

(refer to 3.10 kNkMAIDCapability_WBTuneSunny in MAIDType0016.PDF)

10) WBTuneFlash

Adjust white balance in Flash mode.

(refer to 3.11 kNkMAIDCapability_WBTuneFlash in MAIDType0016.PDF)

11) WBTuneShade

Adjust white balance in Shade mode.

(refer to 3.12 kNkMAIDCapability_WBTuneShade in MAIDType0016.PDF)

12) WBTuneCloudy

Adjust white balance in Cloudy mode.

(refer to 3.13 kNkMAIDCapability_WBTuneCloudy in MAIDType0016.PDF)

13) WBPresetData

Set white balance preset data to camera.

When you set preset data to camera, you have to name the file “PresetData.jpg” and put the file in this application folder in windows, or put the file in home directory in macintosh.

(refer to 3.15 kNkMAIDCapability_WBPresetData in MAIDType0016.PDF)

14) PictureControl

Set and get current Picture Control.

(refer to 3.23 kNkMAIDCapability_PictureControl in MAIDType0016.PDF)

15) PictureControlData

Set and get Picture Control data.

When you set Picture Control data to camera, you have to name the file "PicCtrlData.dat" and put the file in this application folder in windows, or put the file in home directory in macintosh.

(refer to 3.25 kNkMAIDCapability_PictureControlData in MAIDType0016.PDF)

16) GetPicCtrlInfo

Get Picture Control information. The Picture Control information is saved as like "PicCtrlInfo.dat" in application folder in windows, or saved in home directory in macintosh.

(refer to 3.27 kNkMAIDCapability_GetPicCtrlInfo in MAIDType0016.PDF)

17) DeleteCustomPictureControl

Delete Custom Picture Control.

(refer to 3.28 kNkMAIDCapability_DeleteCustomPictureControl in MAIDType0016.PDF)

18) PictureControlDataEx

This will get or set the assigned Picture Control data of the second generation.

(refer to 3.26 kNkMAIDCapability_PictureControlDataEx in MAIDType0016.PDF)

Sub Menu 6

You can read and write settings concerned with Live View.

1) LiveViewProhibit

Get the Live view prohibition condition.

(refer to 3.73 kNkMAIDCapability_LiveViewProhibit in MAIDType0016.PDF)

2) LiveViewStatus

Start or stop Live view, and get current status of Live view.

(refer to 3.72 kNkMAIDCapability_LiveViewStatus in MAIDType0016.PDF)

3) LiveViewImageZoomRate

Set the zoom rate of Live view image.

(refer to 3.74 kNkMAIDCapability_LiveViewImageZoomRate in MAIDType0016.PDF)

4) LiveViewImageSize

This will set the size of “Live View image”.

(refer to 3.75 kNkMAIDCapability_LiveViewImageSize in MAIDType0016.PDF)

5) GetLiveViewImage

Live view data will be got. Display imformation will be saved as like “LiveView001_H.dat” and Live view image will be saved as like “LiveView001.jpg” in application folder in windows, or in home directory in macintosh.

(refer to 3.104 kNkMAIDCapability_GetLiveViewImage in MAIDType0016.PDF)

6) MovRecInCardProhibit

Get the Movie record prohibition condition.

(refer to 3.113 kNkMAIDCapability_MovRecInCardProhibit in MAIDType0016.PDF)

7) MovRecInCardStatus

Start or stop Movie record, and get current status of Movie record.

(refer to 3.112 kNkMAIDCapability_MovRecInCardStatus in MAIDType0016.PDF)

8) MovieScreenSize

This will set the size of movie.

(refer to 3.33 kNkMAIDCapability_MovieScreenSize in MAIDType0016.PDF)

Sub Menu 7

You can read and write a custom setting of source object.

1) EVInterval

Select an interval of exposure value.

(refer to 3.46 kMAIDCapability_EVInterval in MAIDType0016.PDF)

2) BracketingVary

Select a Bracketing variation.

(refer to 3.47 kMAIDCapability_BracketingVary in MAIDType0016.PDF)

Sub Menu 8

You can read and write a setting of image object.

1) IsAlive

Show the status of this image object.

(refer to 9.6 kNkMAIDCapability_IsAlive in MAID3.PDF)

2) Name

Show the name of this image object.

(refer to 9.9 kNkMAIDCapability_Name in MAID3.PDF)

3) StoredBytes

Show the file size of this image file in bytes.

(refer to 9.14 kNkMAIDCapability_StoredBytes in MAID3.PDF)

4) Pixels

Show the image size of this image in pixels.

(refer to 9.40 kNkMAIDCapability_Pixels in MAID3.PDF)

5) RawJpegImageStatus

Show whether this image has raw and jpeg or not.

(refer to 3.102 kNkMAIDCapability_RawJpegImageStatus in MAIDType0016.PDF)

6) Acquire

Get an image file. The image is saved as like “Image001.jpg” in application folder in windows, or saved in home directory in macintosh.

(refer to 9.19 kNkMAIDCapability_Acquire in MAID3.PDF)

Sub Menu 9

You can read and write a setting of movie object.

1) IsAlive

Show the status of this image object.

(refer to 9.6 kNkMAIDCapability_IsAlive in MAID3.PDF)

2) Name

Show the name of this image object.

(refer to 9.9 kNkMAIDCapability_Name in MAID3.PDF)

3) StoredBytes

Show the file size of this image file in bytes.

(refer to 9.14 kNkMAIDCapability_StoredBytes in MAID3.PDF)

4) Pixels

Show the image size of this image in pixels.

(refer to 9.40 kNkMAIDCapability_Pixels in MAID3.PDF)

5) GetVideoImage

Get a movie file. The movie is saved as like “Movie001.jpg” in application folder in windows, or saved in home directory in macintosh.

(refer to 3.105 kNkMAIDCapability_GetVideoImage in MAIDType0016.PDF)

Sub Menu 10

You can read and write a setting of thumbnail object.

1) IsAlive

Show the status of this thumbnail object.

(refer to 9.6 kNkMAIDCapability_IsAlive in MAID3.PDF)

2) Name

Show the name of this thumbnail object.

(refer to 9.9 kNkMAIDCapability_Name in MAID3.PDF)

3) StoredBytes

Show the size of this thumbnail data in bytes.

(refer to 9.14 kNkMAIDCapability_StoredBytes in MAID3.PDF)

4) Pixels

Show the image size of this thumbnail image in pixels.

(refer to 9.40 kNkMAIDCapability_Pixels in MAID3.PDF)

5) Acquire

Get a thumbnail image data. The thumbnail is saved as like “Thumb001.raw” in application folder in windows, or saved in home directory in macintosh.

(refer to 9.19 kNkMAIDCapability_Acquire in MAID3.PDF)

Appendix

About sample program(Macintosh)

- When you switch between release and debug binaries, you change "Build Configuration" of "Edit Scheme" menu from Type0016CtrlSample.xcodeproj.

Restriction

- This sample program doesn't provide all of the capability Type0016 module support. About usage of the other capability, please refer the documet "MAID3Type0016(E).pdf".
- This sample program doesn't care of changes in camera condition. If you want to check the camera status, you should send kNkMAIDCommand_Async to module object periodically. If this command is carried out, change of the state of a camera will be notified by MAIDEventProc.

History

- Rev.2.6 November 14, 2022
  - Files [Windows]...Added “dnssd.dll” and “NkRoyalmile.dll”.

- Rev.2.5 October 29, 2021
  - Environment...Update the environment.

- Rev.2.4 February 26, 2020
  - Environment...Update the environment of Windows.
  - Environment...Update the environment of Macintosh.

- Rev.2.3 November 15, 2019
  - Environment...Update the environment and restrictions of Macintosh.
  - Environment...Update Windows restrictions.

- Rev.2.2 September 28, 2018
  - Environment…Update the environment and restriction of Macintosh.
  - Files…Add necessary file for executing on Macintosh.

- Rev2.1 June 14, 2017
  - Environment…Update the environment and restriction of Macintosh.

- Rev2.0 December 9, 2016
  - Supported camera…D5600 Added.
  - Environment…Update the environment of Macintosh.
  - Sub Menu 1 8)CameraType…Added.
  - Sub Menu 6 8)MovieScreenType…Added.

- Rev1.2 February 29, 2016
  - Environment…Update the environment and restriction of Windows.
  - Sub Menu 1 9)Capture…Update notice.

- Rev1.1 November 13, 2015
  - Environment…Update the environment and restriction of Macintosh.
  - Files…Update the PTP driver of Macintosh.
  - Sub Menu 1 9)Capture…Change description.

- Rev1.0 February 1, 2015 First version